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Software Quality Engineering — SRS Document

## Parcel Dispatch and Tracking System


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# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements of the Parcel Dispatch and Tracking System. This document defines the system's features, expected behaviour, user roles, constraints and acceptance criteria.

It will be used by the development team to build the system and by the testing team to verify that the system works according to the specified requirements.

## 1.2 Scope

The Parcel Dispatch and Tracking System is a web-based application designed for a small courier company. It will help staff register parcels, calculate delivery charges, assign parcels to delivery staff and update delivery statuses.

Customers will be able to check parcel progress through a public tracking page by entering a valid tracking number.

The system aims to reduce manual record-keeping, improve parcel management and make delivery information easier to access.

### Main Features

- User login and role-based access.
- Parcel registration and tracking number generation.
- Automatic delivery charge calculation.
- Assignment of parcels to delivery staff.
- Updating and viewing parcel delivery status.
- Recording failed delivery attempts.
- Managing parcel returns.
- Public parcel tracking for customers.

## 1.3 Definitions and Abbreviations

| Term | Meaning |
|---|---|
| **SRS** | Software Requirements Specification |
| **Admin** | Administrator or authorized courier clerk |
| **Delivery Staff** | Person responsible for delivering parcels |
| **Customer** | Person sending or receiving a parcel |
| **Parcel** | A package registered for delivery |
| **Tracking Number** | Unique code used to identify and track a parcel |
| **FR** | Functional Requirement |
| **NFR** | Non-Functional Requirement |

| Term | Meaning |
|---|---|
| **QA** | Quality Assurance |
| **UI** | User Interface |

### 1.4 Intended Audience

This document is intended for:

- Project developers.
- Software testers and QA members.
- Project documentation members.
- Course instructor and evaluators.
- Other stakeholders involved in the project.

### 1.5 References

- Software Quality Engineering course project brief.
- Requirements and project scope agreed upon by the group.
- Project proposal for the Parcel Dispatch and Tracking System.

## 2. Overall Description

### 2.1 Product Perspective

The Parcel Dispatch and Tracking System will be a standalone web application. It will use a database to store user accounts, parcel details, delivery assignments and shipment history.

The system will have two main areas:

1. **Staff area:** Used by authorized employees to manage parcels and deliveries.
2. **Customer tracking area:** Allows customers to check parcel status without logging in.

### 2.2 Product Functions

The system will provide the following functions:

1. Allow authorized staff to log in.
2. Allow staff to register new parcels.
3. Generate a unique tracking number for each parcel.
4. Calculate delivery charges using predefined pricing rules.
5. Assign parcels to available delivery staff.
6. Update parcel delivery status.
7. Maintain a history of shipment status changes.
8. Record failed delivery attempts and reasons.
9. Process parcel returns.
10. Allow customers to track parcels using tracking numbers.

## 2.3 User Classes and Characteristics

| User | Responsibilities |
|---|---|
| **Administrator / Courier Clerk** | Manage parcels, calculate charges, assign staff and view records. |
| **Delivery Staff** | View assigned parcels and update delivery statuses. |
| **Customer** | Track a parcel using its tracking number. |

The administrator and delivery staff must log in before accessing their respective functions. Customers will not need an account to use the public tracking feature.

## 2.4 Operating Environment

The proposed system will operate in the following environment:

- **Frontend:** HTML, CSS and JavaScript.
- **Backend:** Node.js with Express.js.
- **Database:** MySQL.
- **Testing tools:** Jest and Supertest.
- **Browser:** Modern web browsers such as Google Chrome, Microsoft Edge and Firefox.
- **Operating system:** Windows or another operating system that supports the required development environment.

## 2.5 Design and Implementation Constraints

- The system will be developed as a web application.
- MySQL will be used for database storage.
- Access to staff functions will require authentication.
- Each parcel must have a unique tracking number.
- Parcel status changes must follow defined business rules.
- The system will initially support the features listed in this SRS only.

## 2.6 Assumptions and Dependencies

- Staff members will have access to an internet-connected device and a supported browser.
- Authorized staff accounts will be created by the administrator.
- Delivery staff information will be available before parcel assignment.
- The courier company will provide the pricing rules used by the charge calculator.
- The system will depend on the database being available to store and retrieve records.

# 3. Functional Requirements

Functional requirements describe the specific tasks the system must perform.

## 3.1 User Authentication

| ID | Requirement |
|---|---|
| **FR-01** | The system shall allow authorized staff to log in using a valid username or email and password. |
| **FR-02** | The system shall reject login attempts containing incorrect credentials. |
| **FR-03** | The system shall provide different access permissions for administrators and delivery staff. |
| **FR-04** | The system shall allow logged-in staff to log out securely. |

## 3.2 Parcel Registration

| ID | Requirement |
|---|---|
| **FR-05** | The system shall allow authorized staff to register a new parcel. |
| **FR-06** | The system shall collect the sender's name, receiver's name, contact details, pickup address and delivery address. |
| **FR-07** | The system shall record parcel weight and delivery type. |
| **FR-08** | The system shall generate a unique tracking number for each successfully registered parcel. |
| **FR-09** | The system shall reject parcel registration if required information is missing or invalid. |
| **FR-10** | The system shall allow authorized staff to search for a parcel using its tracking number. |

## 3.3 Delivery Charge Calculation

| ID | Requirement |
|---|---|
| **FR-11** | The system shall calculate delivery charges using the configured pricing rules. |
| **FR-12** | The system shall consider parcel weight, delivery distance and delivery type when calculating the charge, where applicable. |
| **FR-13** | The system shall display the calculated delivery charge before the parcel registration is confirmed. |
| **FR-14** | The system shall reject invalid values such as zero or negative parcel weight. |

## 3.4 Staff Assignment

| ID | Requirement |
|---|---|
| **FR-15** | The system shall allow the administrator or authorized clerk to assign a parcel to delivery staff. |
| **FR-16** | The system shall display available delivery staff for assignment. |
| **FR-17** | The system shall store the assigned staff member's details with the parcel record. |
| **FR-18** | The system shall prevent dispatch of a parcel that has not been assigned to delivery staff. |

## 3.5 Shipment Tracking and Status Management

| ID | Requirement |
|---|---|
| **FR-19** | The system shall allow authorized staff to update parcel delivery status. |
| **FR-20** | The system shall support the following parcel statuses:<br>• Booked<br>• Assigned<br>• Dispatched<br>• Out for Delivery<br>• Delivered<br>• Delivery Failed<br>• Return Initiated<br>• Return in Transit<br>• Returned to Sender |
| **FR-21** | The system shall record the date and time of each status update. |
| **FR-22** | The system shall maintain a shipment history for each parcel. |
| **FR-23** | The system shall prevent invalid status transitions according to the defined business rules. |

## 3.6 Failed Delivery and Returns

| ID | Requirement |
|---|---|
| **FR-24** | The system shall allow delivery staff to record a failed delivery attempt. |
| **FR-25** | The system shall require a reason when a delivery attempt is marked as failed. |
| **FR-26** | The system shall record the number of delivery attempts made for a parcel. |
| **FR-27** | The system shall allow authorized staff to initiate a parcel return according to the company's return policy. |
| **FR-28** | The system shall update the parcel status during the return process. |

### 3.7 Customer Tracking

| ID | Requirement |
|---|---|
| **FR-29** | The system shall provide a public tracking page. |
| **FR-30** | The customer shall be able to enter a tracking number to search for a parcel. |
| **FR-31** | The system shall display the current status of a parcel when a valid tracking number is entered. |
| **FR-32** | The system shall display the parcel's shipment status history to the customer. |
| **FR-33** | The system shall display an appropriate message when an invalid tracking number is entered. |
| **FR-34** | The public tracking page shall not expose private staff account information. |
| **FR-35** | The system shall display relevant delivery information without exposing unnecessary personal information. |

## 4. Non-Functional Requirements

Non-functional requirements describe the quality standards and conditions under which the system should operate.

### 4.1 Usability

| ID | Requirement |
|---|---|
| **NFR-01** | The system shall provide a simple and understandable user interface. |
| **NFR-02** | The system shall display clear error messages when users enter invalid information. |
| **NFR-03** | The system shall provide clear labels and instructions for important fields and operations. |
| **NFR-04** | The system shall allow customers to track parcels without creating an account. |

### 4.2 Performance

| ID | Requirement |
|---|---|
| **NFR-05** | Under the agreed test environment and test load, at least 95% of normal system requests should complete within 2 seconds. |
| **NFR-06** | The system shall display parcel tracking results within an acceptable response time under the agreed test load. |
| **NFR-07** | The system shall support multiple staff members using the application at the same time, within the capacity defined for project testing. |

## 4.3 Reliability

| ID | Requirement |
|---|---|
| **NFR-08** | The system shall store parcel information accurately and prevent incomplete records from being saved. |
| **NFR-09** | The system shall preserve shipment history when a parcel status is updated. |
| **NFR-10** | The system shall display an appropriate error message if the database or a required service is unavailable. |
| **NFR-11** | The system shall prevent duplicate tracking numbers. |

## 4.4 Security

| ID | Requirement |
|---|---|
| **NFR-12** | The system shall require authentication before allowing access to staff functions. |
| **NFR-13** | The system shall restrict system functions according to user roles. |
| **NFR-14** | The system shall store passwords securely using an appropriate password-hashing method. |
| **NFR-15** | The system shall validate user input to reduce common security risks. |
| **NFR-16** | The public tracking page shall not allow customers to modify parcel records. |

## 4.5 Maintainability

| ID | Requirement |
|---|---|
| **NFR-17** | The system shall use organized and understandable source code. |
| **NFR-18** | The system shall separate frontend, backend and database responsibilities. |
| **NFR-19** | The system shall allow future changes to delivery pricing rules without requiring a complete redesign. |

## 4.6 Compatibility

| ID | Requirement |
|---|---|
| **NFR-20** | The system shall work on commonly used modern web browsers. |
| **NFR-21** | The interface shall remain usable on desktop and laptop screen sizes. |

## 4.7 Data Integrity

| ID | Requirement |
|---|---|
| **NFR-22** | Each parcel shall have a unique tracking number. |
| **NFR-23** | Required parcel information shall be validated before saving. |

| ID | Requirement |
|---|---|
| **NFR-24** | Each shipment status update shall be associated with the correct parcel. |

# 5. Use-Case Diagram and Use Cases

## 5.1 Actors

The system has three main actors:

1. **Administrator / Courier Clerk:** Registers parcels, calculates charges, assigns delivery staff and manages parcel records.
2. **Delivery Staff:** Views assigned parcels and updates delivery information.
3. **Customer:** Tracks parcels using a tracking number.

## 5.2 Use-Case Diagram

The following diagram represents the main interactions between users and the system.

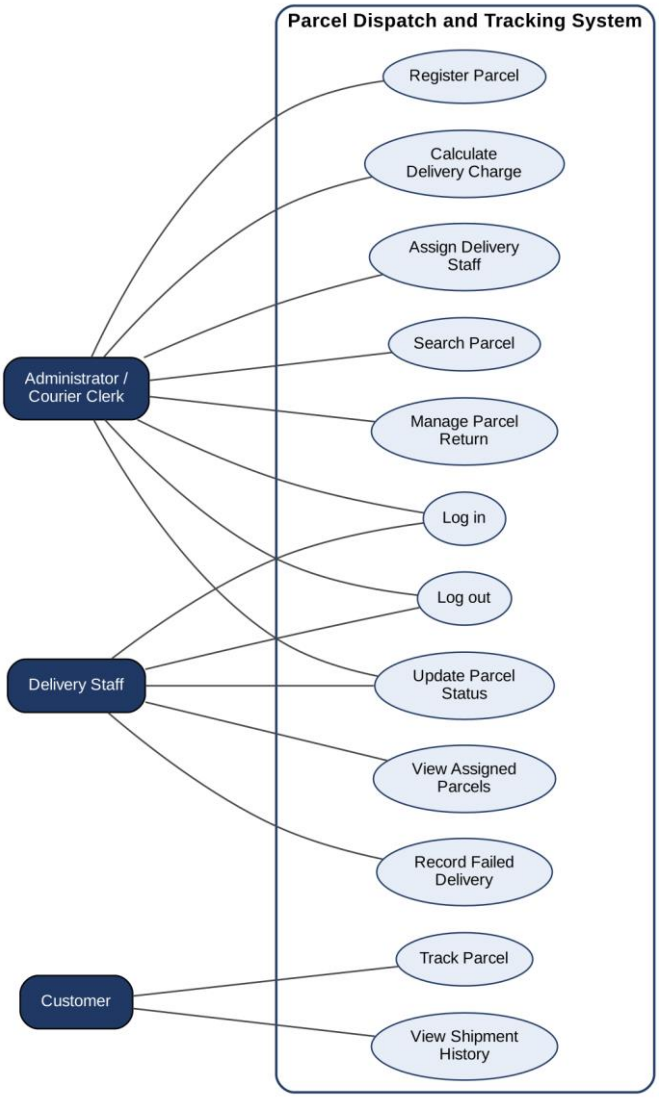



*Figure 1: Use-case diagram of the Parcel Dispatch and Tracking System*

## 5.3 Use-Case List

| Use Case ID | Use Case | Actor |
|---|---|---|
| **UC-01** | Log in | Administrator, Delivery Staff |
| **UC-02** | Register Parcel | Administrator / Courier Clerk |
| **UC-03** | Calculate Delivery Charge | Administrator / Courier Clerk |
| **UC-04** | Assign Delivery Staff | Administrator / Courier Clerk |
| **UC-05** | Search Parcel | Administrator / Courier Clerk |
| **UC-06** | View Assigned Parcels | Delivery Staff |
| **UC-07** | Update Parcel Status | Delivery Staff |
| **UC-08** | Record Failed Delivery | Delivery Staff |
| **UC-09** | Manage Parcel Return | Administrator / Courier Clerk |
| **UC-10** | Track Parcel | Customer |
| **UC-11** | View Shipment History | Customer |

## 5.4 Detailed Use Case: UC-02 – Register Parcel

| | |
|---|---|
| **Use Case ID** | UC-02 |
| **Use Case Name** | Register Parcel |
| **Primary Actor** | Administrator / Courier Clerk |
| **Preconditions** | • The staff member must be logged in.<br>• The staff member must have permission to register parcels. |
| **Main Flow** | 1. The staff member opens the parcel registration page.<br>2. The system displays the registration form.<br>3. The staff member enters sender and receiver details.<br>4. The staff member enters pickup and delivery addresses.<br>5. The staff member enters parcel weight and delivery type.<br>6. The system validates the entered information.<br>7. The system calculates the delivery charge.<br>8. The system generates a unique tracking number.<br>9. The system saves the parcel record.<br>10. The system displays a registration confirmation and tracking number. |
| **Alternative Flow** | • If required information is missing, the system displays an error message.<br>• If parcel weight is invalid, the system rejects the entry and requests a valid value.<br>• If the database is unavailable, the system displays an error and does not show a successful registration message. |
| **Postconditions** | • A valid parcel record is saved.<br>• A unique tracking number is assigned.<br>• The initial parcel status is set to Booked. |

### 5.5 Detailed Use Case: UC-10 – Track Parcel

| | |
|---|---|
| **Use Case ID** | UC-10 |
| **Use Case Name** | Track Parcel |
| **Primary Actor** | Customer |
| **Preconditions** | • The customer must have a valid tracking number.<br>• The public tracking page must be available. |
| **Main Flow** | 1. The customer opens the tracking page.<br>2. The system displays a tracking number field.<br>3. The customer enters the tracking number.<br>4. The customer selects the Track option.<br>5. The system searches the parcel database.<br>6. The system displays the current parcel status.<br>7. The system displays the available shipment history. |
| **Alternative Flow** | • If the tracking number is invalid, the system displays an appropriate error message.<br>• If no parcel is found, the system informs the customer that no matching record exists. |
| **Postconditions** | • The customer can view the available tracking information.<br>• No parcel information is changed by the tracking operation. |

## 6. Business Rules

The following rules define how the system should handle important operations.

| ID | Business Rule |
|---|---|
| **BR-01** | Every registered parcel must have a unique tracking number. |
| **BR-02** | Parcel registration requires all mandatory sender, receiver and parcel details. |
| **BR-03** | Parcel weight must be greater than zero. |
| **BR-04** | Delivery charges must be calculated using the pricing rules approved by the courier company. |
| **BR-05** | A parcel must be assigned to delivery staff before it can be dispatched. |
| **BR-06** | A parcel marked as Delivered cannot be returned to an earlier delivery status through an ordinary status update. |
| **BR-07** | A failed delivery attempt must include a reason. |
| **BR-08** | Every status change must be recorded in the shipment history with its date and time. |
| **BR-09** | Only authorized staff may change parcel information or delivery status. |
| **BR-10** | Customers may view parcel tracking information but cannot modify parcel records. |
| **BR-11** | Parcel returns must follow the return procedure approved by the courier company. |

# 7. External Interface Requirements

## 7.1 User Interface Requirements

The system shall provide:

- A staff login page.
- A staff dashboard.
- A parcel registration form.
- A delivery charge display.
- A staff assignment page.
- A parcel search page.
- A delivery status update page.
- A public customer tracking page.

## 7.2 Software Interface Requirements

The system will use:

- Node.js and Express.js for backend processing.
- HTML, CSS and JavaScript for the frontend.
- MySQL for storing application data.
- Jest and Supertest for automated testing.

## 7.3 Database Interface Requirements

The database shall store the following main types of information:

- User accounts and roles.
- Parcel details.
- Delivery staff assignments.
- Shipment status history.
- Delivery attempts.

## 7.4 Communication Interface Requirements

The application shall communicate between the frontend and backend using HTTP requests. Database operations shall be handled by the backend.

# 8. Requirements Traceability Matrix

The traceability matrix connects requirements with test cases. It helps the QA team confirm that important requirements are tested.

| Requirement ID | Requirement Description | Test Case ID |
|---|---|---|
| **FR-01** | Staff login with valid credentials | TC-01 |
| **FR-02** | Reject incorrect login credentials | TC-02 |
| **FR-05** | Register a new parcel | TC-03 |
| **FR-09** | Reject missing or invalid parcel information | TC-04 |
| **FR-08** | Generate a unique tracking number | TC-05 |
| **FR-11** | Calculate delivery charges correctly | TC-06 |
| **FR-14** | Reject zero or negative parcel weight | TC-07 |
| **FR-15** | Assign parcel to delivery staff | TC-08 |
| **FR-18** | Prevent dispatch of an unassigned parcel | TC-09 |
| **FR-19** | Update parcel delivery status | TC-10 |
| **FR-23** | Prevent invalid status transitions | TC-11 |
| **FR-25** | Require a reason for failed delivery | TC-12 |
| **FR-27** | Initiate a parcel return | TC-13 |
| **FR-30** | Search using a valid tracking number | TC-14 |
| **FR-33** | Display an error for an invalid tracking number | TC-15 |
| **NFR-05** | Meet the agreed response-time target | TC-16 |
| **NFR-13** | Restrict access according to user roles | TC-17 |
| **NFR-14** | Store passwords securely | TC-18 |

# 9. Acceptance Criteria

Acceptance criteria define the conditions that must be met for important requirements to be considered successfully implemented.

| Feature | Acceptance Criteria |
|---|---|
| **Staff Login** | A user with valid credentials can log in, while invalid credentials are rejected. |
| **Parcel Registration** | A parcel with all valid required information is saved successfully and receives a unique tracking number. |
| **Input Validation** | Missing required fields and invalid parcel weights are rejected with clear error messages. |

| Feature | Acceptance Criteria |
|---|---|
| **Charge Calculation** | The displayed delivery charge matches the configured pricing formula for the entered values. |
| **Staff Assignment** | An authorized staff member can assign a parcel to a valid delivery staff member. |
| **Dispatch** | The system does not allow an unassigned parcel to be dispatched. |
| **Status Management** | Valid status changes are saved and recorded in shipment history. |
| **Failed Delivery** | A failed delivery attempt cannot be saved without a reason. |
| **Parcel Tracking** | A valid tracking number displays the correct parcel status and available history. |
| **Invalid Tracking** | An unknown tracking number produces a clear message without exposing unrelated parcel data. |
| **Role Security** | Delivery staff cannot access administrator-only functions. |
| **Performance** | At least 95% of normal requests complete within 2 seconds under the agreed test environment and load. |

## 10. Future Enhancements

The following features may be considered for future versions but are not included in the current project scope:

- Live GPS parcel tracking.
- SMS and email delivery notifications.
- Online payment integration.
- Route optimization.
- Integration with external courier services.
- Customer account registration.

## 11. Conclusion

This Software Requirements Specification defines the main functional and non-functional requirements of the Parcel Dispatch and Tracking System. It describes the expected system behaviour, user roles, business rules, use cases, acceptance criteria and requirement-to-test-case traceability.

This document will guide the development and testing teams throughout the project. It will also provide a basis for evaluating whether the completed system meets the requirements established for the project.

# — End of Document —